Uppsala University

Bayesian Statistics and Data Analysis

Mini-Project Instructions

Contents

1 Project Instructions

The last two weeks will focus on a course project where 2-3 students either choose data and do a Bayesian analysis of a real-world dataset, or choose the paper replication track described below.

Requirements for the projects are:

- The ordinary project track should be a Bayesian Data Analysis using Stan (you can use `brms` or `rstanarm` if you like). Although, this is easier and will be considered when grading, i.e. it will be harder to get a VG in the mini-project.
- The paper replication track should reproduce a published statistical analysis by implementing its data and model in Stan and preparing the result for inclusion in `posteriordb` (inclusion in `posteriordb` is optional).
- Real data should be used (see below for details).
- In the ordinary project track, at least two different models should be estimated and compared.
- In the paper replication track, one faithful implementation of a sufficiently interesting or complex model is enough (i.e. a model not used in the course). The comparison should instead be between the published results and the Stan implementation.

Tip! Write the report so you can add your final report as an example of work you have done to future potential employers.

For PhD students: You can choose to make a small project related to your research interest instead. Although, it should still be a 4 page paper output.

1.1 Suggested Reading/Video material

The project will be a small practical exercise in Bayesian data analysis. To get some inspiration, see the Stan YouTube channel [here] or the Stan User Guide [here].

For guidance on communicating posterior uncertainty and predictive distributions in figures, see Matthew Kay's StanCon 2026 lecture Systematic uncertainty visualization design.

For the paper replication track, see the `posteriordb` repository [here] and the guide for adding content using R [here].

1.2 Project Group and Expected Workload

It is possible to have only one student in a group, although this is not recommended. One student group will, in practice, mean additional work due to the requirements of the project.

The project is expected to take 40h per student in the group. Hence a 3 group project should be the equivalent of a 120h project.

One student (first author) is the corresponding author. Thats the one that submit the paper on studium.

1.3 Data Sets and Methods Recommendations

We recommend that you find a dataset you are interested in using yourself, ideally in a field you find interesting. Feel free to discuss potential projects with the teacher.

If you have a hard time finding a dataset to use, there are a lot of available datasets (and problems) at:

- The UCI Machine Learning repository: [here]
- The machine learning competition site Kaggle: [here]

Hint! Use a data set with a natural grouping (e.g. patients in hospitals, car emissions by different brands, data with a time dimension etc) since that will help you to use ideas of hiearchical modeling from later in the course.

The following data sets should not be used in the project:

- Titanic (R data set)
- mtcars (R data set)

Also, two different groups are not allowed to use the same dataset.

For the paper replication track, students should instead find a published statistical analysis where the data and the model description are open. The original analysis can be frequentist or Bayesian, but the submitted implementation must be in Stan and must use proper priors. The students should be ok with releasing the stan code openly. Two different groups are not allowed to use the same paper/model/data combination.

1.4 Project Proposal (Part 1 and Part 2)

Students need to turn in a one-page project proposal and data description by and get approval for the proposed project. The Project proposal should follow the ICML paper format that can be found [here]. The ICML format is also available in overleaf here: [here]. We recommend using Overleaf for writing the proposal.

Note! You should also follow the formating instructions in the ICML template, e.g. regarding Tables, Figures etc.

The project proposal should include all the group members names!

In the ordinary Project proposal part 1, the following parts should be included. See Project report below on the details on each section.

1. Title
2. Abstract (1 paragraph)

3. Introduction (roughly 0.5 page)
4. Data (roughly 0.5 page)
5. Models (roughly 0.25 page) *Note!* For proposal part 1, you only need the most simple model to model the actual data.

For Project Proposal part 2, you need to add the following to your report:

1. Stan code in the appendix for the first proposed model (from part 1)
2. The first model estimated and added to the Results section of the project proposal.
3. Address comments you got on part 1.

Paper replication project proposal For the paper replication track, the Project proposal part 1 should instead include:

1. Title
2. Abstract (1 paragraph)
3. Introduction to the paper and statistical problem (roughly 0.5 page)
4. Paper, data, and model sources, including links and (potential) license information
5. A short motivation for why the paper/model is interesting
6. Description of the original statistical model
7. A statement of whether the original analysis is Bayesian or frequentist. If it is frequentist, explain the Bayesian version to be implemented in Stan, including the choice of proper priors.
8. A list of the main results from the paper that will be checked against the Stan implementation
9. A short assessment of whether the data and model are suitable for `posteriordb` (see the `posteriordb` paper)

For Project Proposal part 2 in the paper replication track, you need to add:

1. Stan code in the appendix for the proposed model
2. A first fit of the Stan model, including basic diagnostics
3. A preliminary comparison with at least one result from the original paper
4. A short description of the planned `posteriordb` data, model, and posterior objects
5. Address comments you got on part 1

1.5 Project Report

The Project outcome is a report in the ICML paper format that can be found [here]. The ICML format is also available in overleaf here: [here]. We recommend using Overleaf for writing the report.

Note! You should also follow the formating instructions in the ICML template, e.g. regarding Tables, Figures etc.

The paper should consist of *between three and a half (3.5) and four (4) pages*, excluding references and eventual appendices. Write the report as you would do in a real situation, i.e. do not refer to the paper as a "mini-project" or similar.

The project report should include all the group members names!

The paper should include the following parts/sections:

1. Title
   - The title should describe the problem and be like a real article title, i.e. don't write "Mini-project:" or similar in the title.
2. Abstract (1 paragraph)
3. Introduction (roughly 0.5 page) *Note!* The introduction should be as an introduction to an article, i.e. describe the problem and include at least one reference.
   - Description of the problem.
4. Data (roughly 0.5 page)
   - Description of the data, e.g. the number of observations, the number of groups (if you intend to use a hierarchical model), what is the dependent variable etc.
5. Models and methods (roughly 0.5 page)
   - Description of the models *Note 1!* Only Bayesian Models are allowed, i.e. with proper priors. *Note 2!* Stan code for all models should be included as an Appendix.
   - Description of how the models were compared (LOO/WAIC)
6. Results (roughly 1.5-2 pages)
   - Results of the different models.
   - Estimated parameters of interest together with their posterior uncertainty.
   - Which model does seem to work the best, and why?
7. Conclusions (roughly 0.5-1 pages)
   - Conclusions from the results.
   - Discussion of problems and potential improvements and other models
8. Acknowledgements (optional)

- If you are willing to let me use your project report as an example in the next course, please state that in the Acknowledgements. Just add the following sentence: "We hereby grant our consent for the utilization of this project report as a reference material within the context of future editions of the course."
- Other people you might want to thank.

9. Appendix

   - Stan code for all models used in the report should be included in the appendix
   - Other results, Figures etc that you think might be of interest to some readers.

Paper replication project report For the paper replication track, the report should have the same length and format, but the content should be adapted to the reproduction task. It should include:

1. Introduction: describe the original paper, the scientific question, and why the model is interesting.
2. Data: describe the original data, preprocessing, and how the data were converted to Stan format.
3. Model and methods: describe the original model, the Stan implementation, priors, and any deviations from the paper.
4. Computation: report diagnostics such as $\hat{R}$, effective sample sizes, divergences, and relevant computational issues.
5. Reproduction results: compare selected results from the paper with the Stan results and discuss agreement or disagreement.
6. `posteriordb` contribution (in appendix): describe the data object, Stan model, posterior object, model info object, and whether a pull request was submitted or the files are ready for submission.

For this track, model comparison using LOO/WAIC is not required unless it is relevant for reproducing the paper. The central comparison is between the published analysis and the Stan implementation - can you replicate the original results?

Additional requirements for the report

1. All Figures using color should have a color-blind friendly color palette. See here and here.
2. Figures showing uncertainty should make clear what uncertainty is being shown, e.g. posterior uncertainty in parameters, posterior predictive uncertainty, model comparison uncertainty, or computational uncertainty.

3. The final report should look like a research paper/article, i.e. try to avoid bullet list and get a good flow in the text. Also do not refer to the report as a mini-project. See it as a real report (or short paper).

4. All model included in the report should have the Stan code included in the appendix.

5. For the paper replication track, the appendix should also include enough information to find or recreate the prepared `posteriordb` files, including data, model, posterior metadata, and any preprocessing code.

6. All legends in figures and text in tables should be the same size as the body text in the report.

7. You should use correct reference systems. A tip is to use
`citet`,
`citep`, and
`bibtex`. This will also simplify your future thesis work.

8. You should include some (or all) estimated parameters together with their uncertainty and $\hat{R}$ and interpret them in the report. You are free to include them in the appendix.

9. You should have addressed all comments you got in the project proposal part 1 and 2.

Additional hints for the report

1. Before you turn in the project, do a language check with a tool such as Grammarly. A project will poor english (errors that would have been spotted with a tool such as Grammarly) will affect your grade downwards.

2. Remember to standardize the independent variables (see Stan recommendations).

1.6 Project Presentations

Presentation details:

- Each project needs to be presented in addition to submitting the mini-project report
- The presentation should be high level, but sufficiently detailed information should be readily available to facilitate answering questions from the audience
- For 1-2 person groups, the presentation should be 10 minutes
- For three-person groups, the presentation should be 15 minutes
- Afterwards, questions will be asked first by other students and then by attending teachers.

Specific presentation recommendations:

- The first slide needs to include the project title and names of the group members.
- The chosen methods(s) should be explained and justified (you are *not* holding this presentation for a hypothetical customer who doesn't care about the details of your methods).
- Big enough font size for text and figure labels should be used to make it easy for the audience to read slides.
- A good rule of thumb is to expect one slide to take 2 minutes to present.
- The last/final slide needs to include your conclusion and names of the group members.

Missing the project presentation

Suppose you cannot attend the presentations due to sickness or similar. In that case, you will have to turn in a video presentation where you present the whole project (i.e. the entire presentation). The teacher then grades this presentation.

To be able to get VG on the project the presentation needs to be turned in the day before the presentation at 23.59 the latest.

1.7 Project Grading

Below are the criteria used when grading the mini-projects. Some general comments on grading are:

1. The more students the higher the quality expected of the project, i.e. a better report is expected from a three-student report than a two-student report.

To pass the report (G), the following criterias should be fulfilled:

1. The report should be turned in and follow the general outline of Section 1.5.
2. The report should follow the additional requirements in Section 1.5.
3. show basic knowledge and understanding of the core concepts of the course by using concepts correctly
4. show an understanding in when certain methods should be used or not, and how
5. for the ordinary project track, use at least two (2) *Bayesian* different models and compare them in a correct way (two linear regression models with different covariates is an example of two different models)
6. for the paper replication track, implement the selected published model and data in Stan, compare the Stan results with selected results from the original paper, and prepare the data, model, posterior, and metadata in a `posteriordb`-compatible form

7. state what has been done in the report with clarity, good english and rigour so it is easy for a reader to understand and follow the paper.

8. correctly use references in the report following the guideline of the template in Section 1.5

To pass the mini-project with distinction (VG), the following criterias also apply in addition to the criteria for passing the report above:

1. show deep knowledge knowledge and understanding of the core concepts and how to adapt them in a good way to a new situation

2. connect the analysis in the report with other areas in statistics or previous courses taken in the masters program, i.e. not just repeat what has been done in previous labs.

3. use models that has not been part of the BSDA course (but for example presented in other courses)

4. for the paper replication track, choose a substantially interesting or complex model, give a convincing reproduction of the published results, and make a high-quality `posteriordb` contribution with checks passed locally or a pull request prepared